lecture

October 20, 2025

1 approximation resistance

if an algorithm is *approximation resistant*, then it has an upper bound in how well a polynomial time problem can approximate the best solution. beating the upper bound would prove $\mathbb{P} = \mathbb{NP}$

imagine a ranking. you can query an oracle for rankings of two people and it tells you which one is greater. trivial random ordering gives 0.5 worst case.

use SDP

make a graph where if $a < b$, then there is a directed edge w weight 1 from a to b, and a directed edge with weight -1 from b to a. max cut gives a better than optimal configuration for each half.

if the right side of maxcut has more $+$'s going in, then it becomes the right half of the ordering. recurse.